

1. What is a Programming Language?

A programming language is a formal language used to communicate with a computer. It consists of a set of rules (syntax) and instructions that tell the computer what tasks to perform.

Programming languages help developers create software, websites, mobile applications, games, and AI systems.

2. What is a Programming Paradigm?

A programming paradigm is a method or style of writing programs to solve problems efficiently. A **programming paradigm** is a style or approach to writing and organizing programs. It defines how a programmer structures code to solve problems.

There are mainly of 2 types: Procedural oriented programming paradigm, Object oriented programming paradigm

Example

Suppose you want to build a **Student Management System**:

- In **Procedural Programming**, you write functions like `add_student()` and `delete_student()`.
- In **Object-Oriented Programming (OOP)**, you create a Student class with data and methods.

3. Types of programming Paradigm?

Procedural oriented programming paradigm: Procedural Programming is a programming paradigm where the program is divided into **functions (procedures)**. The focus is on the sequence of steps to solve a problem. Ex: C

Object oriented programming paradigm: Object-Oriented Programming is a programming paradigm that organizes programs using **classes** and **objects**. It combines **data (attributes)** and **functions (methods)** into a single unit called an object. Ex: Java

4. Why is Python Called a Multi-Paradigm Language?

Python is called a multi-paradigm language because it supports different programming styles. A developer can write code using procedural programming, object-oriented programming, or functional programming, depending on the project requirements. This flexibility makes Python suitable for many types of applications.

5. Why should we learn Python?

Python is a great language for beginners because it is simple, easy to understand, and easy to write. It is also one of the most popular programming languages and is used in many fields like web development, data science, artificial intelligence, machine learning, automation, and cybersecurity. Learning Python opens up many career opportunities in the software industry.